

Movement analysis report

RehabMotion AI | Video-based movement analysis for rehabilitation

Source file	sample_squat.mp4
Exercise	Squat
Analyzed side	Right
Generated	2026-07-16 12:30 UTC

This prototype is for educational and R&D purposes only. It is not a certified medical device and should not be used for diagnosis, treatment decisions or clinical monitoring.

Analysis overview

Detected repetitions	Knee ROM	Hip ROM
2	84.0 deg	64.0 deg
Mean rep duration	Max trunk lean	Tempo regularity
3.00 s	23.0 deg	High

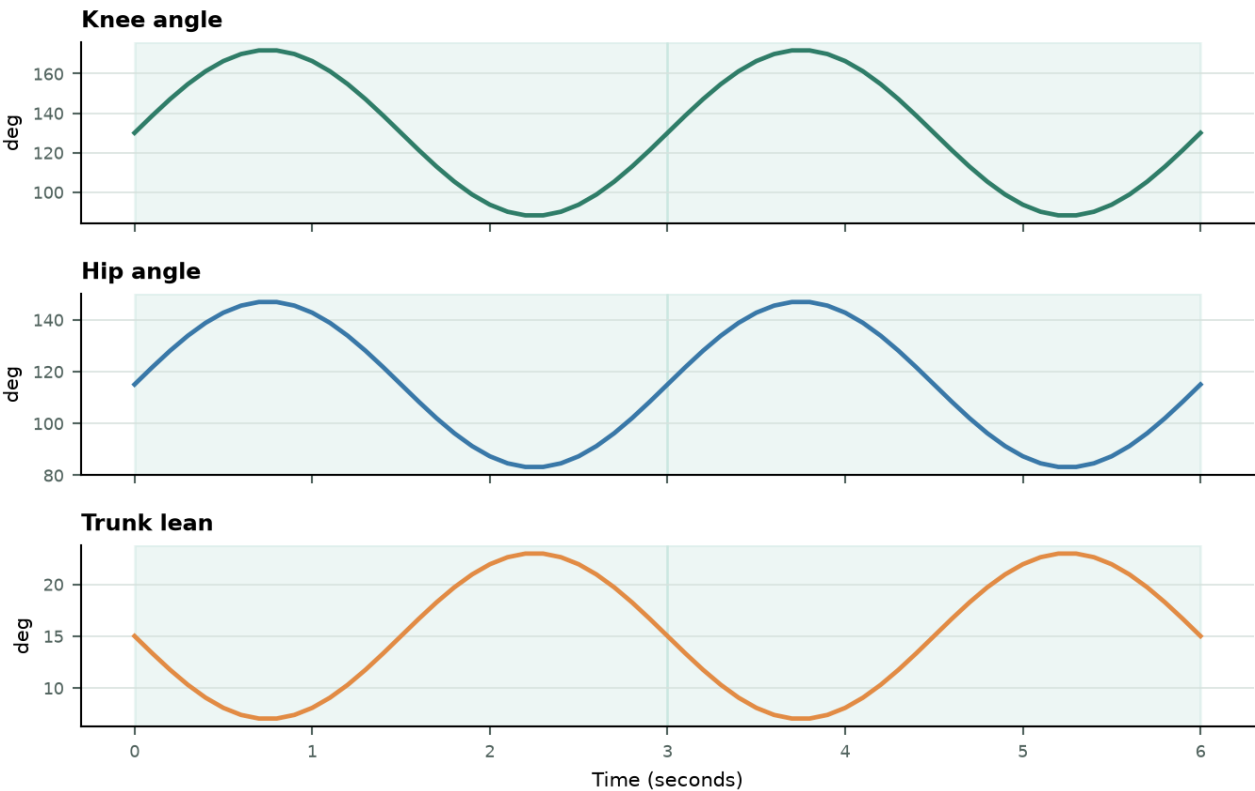
Video and data quality

Video duration	6.00 s
Video rate	30.00 FPS
Frame count	180
Resolution	1280 x 720
Pose detected	98.0%
Bilateral reliability	84.0%
Mean visibility	0.91
Usable kinematics	95.0%

Joint-angle curves

Smoothed 2D estimates. Shaded intervals mark complete detected repetitions. Missing or low-confidence samples are not interpolated across long gaps.

Smoothed 2D kinematics - Right side



Kinematic summary

Knee angle range	88.0 to 172.0 deg
Knee ROM	84.0 deg
Hip angle range	74.0 to 138.0 deg
Hip ROM	64.0 deg
Mean trunk lean	15.4 deg
Max trunk lean	23.0 deg
Mean knee asymmetry	3.2 deg
Max knee asymmetry	7.1 deg

Repetition analysis

Only complete cycles crossing both adaptive thresholds are counted. Partial movements are ignored.

Rep	Duration (s)	Descent (s)	Ascent (s)	Knee ROM (deg)	Hip ROM (deg)	Max trunk (deg)	Mean asym. (deg)
1	3.00	1.50	1.50	84.00	64.00	23.00	3.10
2	3.00	1.50	1.50	82.00	62.00	22.00	3.40

Detection details

Signal excursion	84.0 deg
Start / return threshold	151.0 deg
Turning threshold	102.0 deg
Duration variability	0.0%

Interpretation and limitations

- Results are single-camera 2D estimates and depend on camera placement.
- Occlusion, loose clothing, lighting and out-of-plane motion reduce accuracy.
- ROM values are not equivalent to clinical goniometry or 3D motion capture.
- Tempo and repetition thresholds are heuristics requiring visual review.
- Left-right asymmetry requires both sides to be visible at the same time.

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